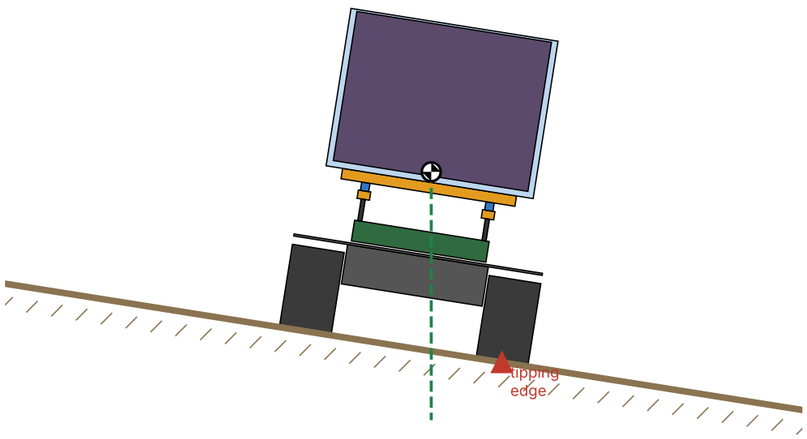
BINBOT — STABILITY LIMITS, SLOPES UP TO 30° (top: full bin 500 kg raised · bottom: 400 kg, deck lowered)

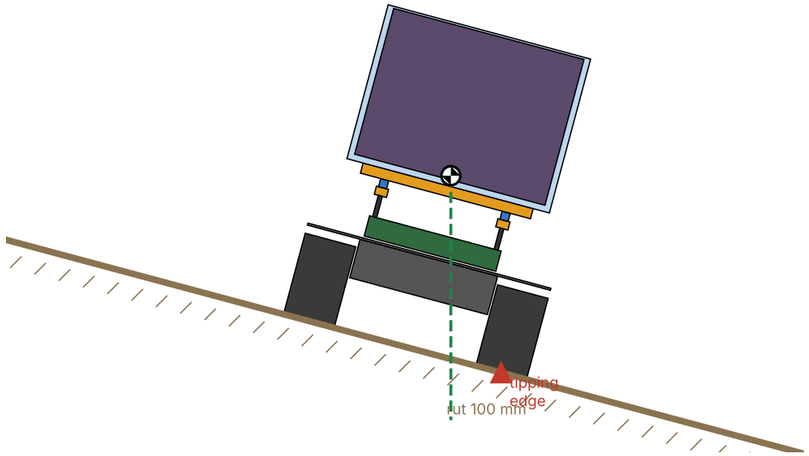
A CROSS-SLOPE LIMIT 9° (driving)

OK: CoG well inside.
Alarm 7°, auto-stop 12°.



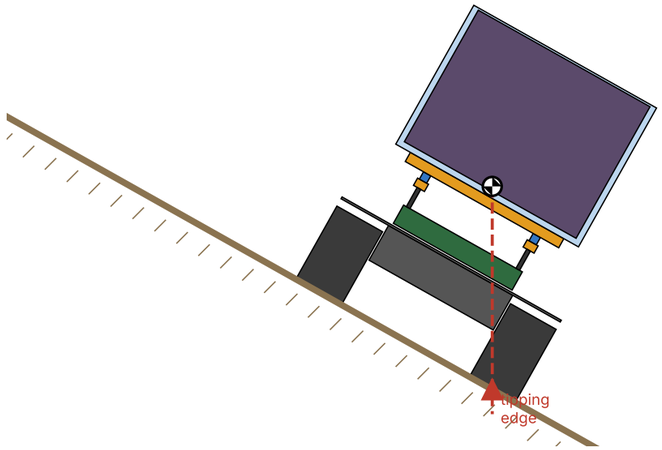
B ROAD: 9° CAMBER + 100 mm RUT = 15°

OK: still 14° from tipping.
Ruts are why the limit is far below the tip angle.



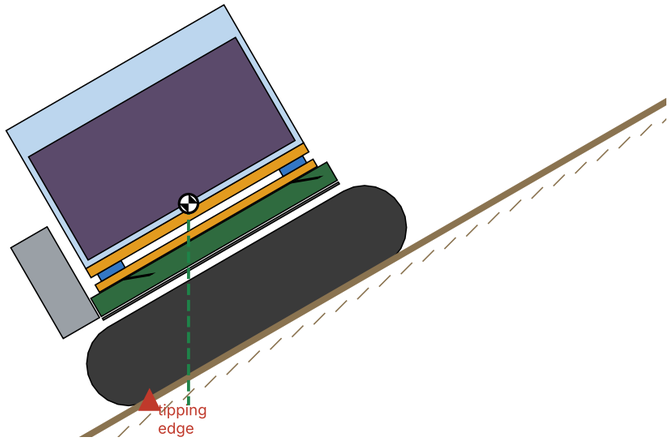
C SIDEWAYS TIP POINT 29° — NEVER

CoG exactly over the edge:
the robot rolls over.



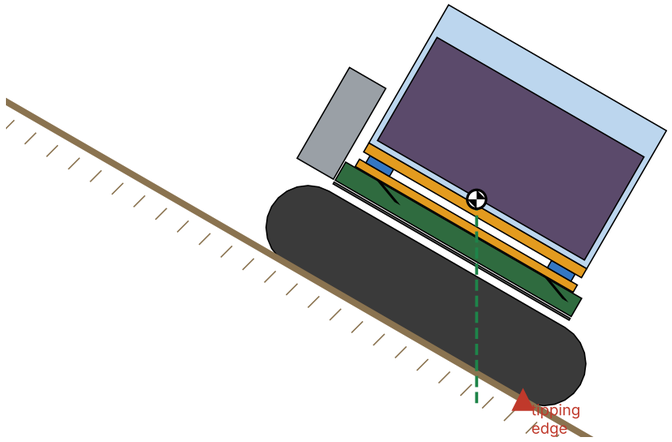
D CLIMBING 30°, 400 kg, deck low

Tips backwards at 41°: 11° margin.
Grip needed 0.70 × weight: DRY ONLY.
Better: climb empty, carry full bins downhill.



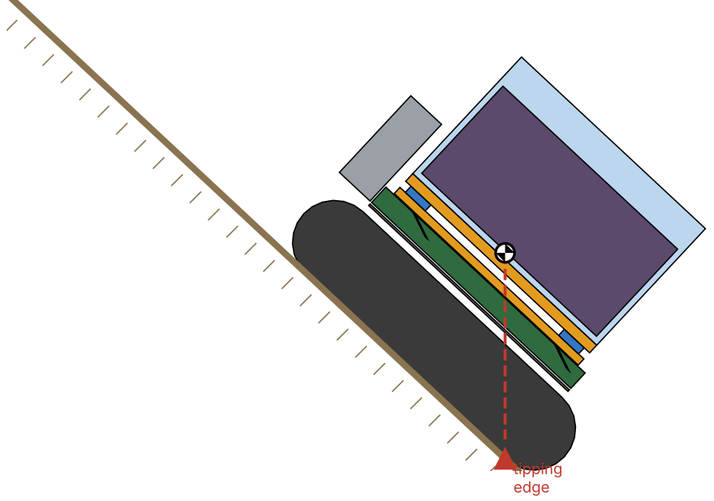
E GOING DOWN 30°, 400 kg, deck low

Tips forwards at 43°: 13° margin.
Brake gently (0.5 m/s² ≈ 3°).
Full 500 kg raised would leave only 6°: NOT allowed.



F FORWARD TIP POINT 43° — NEVER

CoG over the front edge
of the track contact.



Load state	CoG height	Tip sideways	Tip fwd / back	Cross-slope limit	Up/down limit
Full bin 500 kg, deck RAISED	845 mm	29°	36° / 39°	9°	25°
Full bin 500 kg, deck LOWERED (road mode)	790 mm	31°	40° / 39°	9°	28°
400 kg olives, deck LOWERED (30° routes)	719 mm	33°	43° / 41°	10°	30°
Empty bin	457 mm	46°	56° / 54°	14°	30°
No bin	393 mm	50°	60° / 57°	15°	30°

RULES FOR 30° SLOPES
- Deck LOWERED whenever the slope is > 15° (road mode, 8 kN actuators)
- Bins on 30° routes filled to ≤ 400 kg (software sets the target)
- Base is UPHILL: full bins climb → dry only, rear sprags armed (DWG-005)
- GRIP: standing still on 30° needs 0.58 × weight, climbing 0.70.
  Wet clay gives 0.4–0.6 → the robot can SLIDE even braked.
  30° routes are DRY-ONLY; blocked in wet-day mode or on slip.

Masses are estimates: weigh the robot once built and
re-run cad/stability.py; the software limits use these values.

Operating limits: sideways ≈ 30 % of the tip angle (ruts, bumps, turning, soft ground); up/down = tip angle − 11°
(3° braking + 8° bumps), capped at 20° / 30°. Sideways tipping edge taken at the track centre line (soft ground).

DWG 004 REV C stability 2026-10-28
PRELIMINARY — masses estimated, base data assumed